

Testing and Commissioning Protocol for Utility-Scale Solar Photovoltaic Power Plants: The HELIOS Project

Educational & Safety Notice: The technical procedures, sequences, and configurations described in this report outline general industry standards and engineering philosophies. For high-voltage operations, switching programs, and safety-critical commissioning execution, operators must consult qualified professionals, adhere to original equipment manufacturer (OEM) manuals, and comply with all local statutory safety and electrical regulations.

1. Testing & Commissioning Overview & Philosophy

The Testing and Commissioning (T&C) phase of a utility-scale solar photovoltaic (PV) power plant represents the highly structured transition from a static construction site to a dynamic, grid-synchronized generation asset. For a 100 MW facility such as the HELIOS project, which interfaces directly with the UzbekEnergo 220kV transmission grid, the commissioning philosophy is predicated on a phased, bottom-up validation of generation components coupled with a strictly controlled top-down energisation sequence from the utility grid.

Commissioning Philosophy and Sequencing

The fundamental philosophy dictates that no electrical subsystem is subjected to live voltage until its physical integrity, mechanical operation, and static electrical parameters are exhaustively proven. The process is divided into three distinct phases:

1. **Pre-Commissioning (Cold Testing):** This phase encompasses all de-energized tests intended to verify that the installation matches the approved design and complies with international standards. It involves visual inspections, torque verifications, mechanical interlock testing, continuity checks, and crucial insulation resistance (IR) and dielectric withstand measurements.
2. **Commissioning (Live Testing and Energisation):** This phase marks the initial application of voltage. The sequence is rigidly "top-down." The utility 220kV grid is introduced to the Gas Insulated Switchgear (GIS), followed by the 33/220kV step-up transformer, the 33kV medium voltage (MV) ring main units (RMU), the inverter stations, and finally the direct current (DC) solar arrays.
3. **Performance Testing:** Once the plant is fully energized and the inverters are synchronized, the facility undergoes quantitative evaluations. These tests measure the energy yield, the Performance Ratio (PR), and dynamic grid code compliance to ensure the plant meets contractual obligations and utility requirements.

The rationale for the strict top-down energisation sequence is rooted in protective relay coordination. Reversing this sequence or energising subsystems out of order risks exposing generation equipment to severe electrical faults without active upstream protection. By energising the substation first, the primary differential and distance relays are placed into active service, establishing a protected envelope before any downstream MV or LV circuits are made live.

The Role and Authority of the T&C Manager

Upon the issuance of Mechanical Completion (MC) certificates, jurisdictional control of the plant transfers from the Construction Manager to the T&C Manager. The T&C Manager operates as the ultimate technical authority on site. They exercise absolute control over the Lock-Out/Tag-Out (LOTO) system, authorize all switching programs, and possess the authority to halt any operation that threatens personnel safety or equipment integrity. Furthermore, the T&C Manager serves as the primary technical liaison with the utility dispatcher (UzbekEnergo), the owner's engineers, and the original equipment manufacturers.

The Commissioning Plan Structure

A comprehensive Commissioning Plan serves as the master roadmap for the entire operation. It provides the framework for safe and systematic execution.

<u>Section</u>	<u>Description</u>	<u>Key Contents</u>
<u>1.0 Introduction & Scope</u>	<u>Project boundaries</u>	<u>100 MW PV Plant, 33/220kV Substation, relevant IEC standards.</u>
<u>2.0 HSE & LOTO Management</u>	<u>Safety protocols</u>	<u>Permit to Work (PTW) system. Sanction for Test (SFT) procedures, isolation boundary management.</u>
<u>3.0 Organization & Roles</u>	<u>T&C Hierarchy</u>	<u>Defines roles for the T&C Manager, Lead Protection Engineer, SCADA Engineer, and Utility Interfaces.</u>

<u>4.0 Pre-Commissioning</u>	<u>Cold testing protocols</u>	<u>Detailed methodologies for DC string tests, Transformer TTR/DGA, RMU Hi-pot testing.</u>
<u>5.0 Energisation Sequence</u>	<u>Hot commissioning</u>	<u>Step-by-step switching programs from 220kV dead bus to live inverter operation.</u>
<u>6.0 Grid Compliance & PR</u>	<u>Utility requirements</u>	<u>Methodologies for Performance Ratio (PR) tests, Fault Ride-Through (FRT), and reactive power validation.</u>
<u>7.0 Handover</u>	<u>Documentation</u>	<u>turnover packages, Provisional Acceptance Certificate (PAC) criteria, and O&M training.</u>

2. DC System Pre-Commissioning

The DC infrastructure, comprising thousands of PV modules, string cables, combiner boxes, and main DC feeders, requires meticulous validation before connection to the inverter. International standard IEC 62446-1 dictates the minimum requirements for the testing of PV systems.

Pre-Commissioning Checks for DC Strings

Prior to inverter energisation, a rigorous suite of tests must be executed for every string. These tests verify both performance and safety boundaries.

<u>Test Description</u>	<u>Methodology</u>	<u>Acceptance Criteria</u>

<u>Polarity Check</u>	<u>Conducted at the combiner box using a digital multimeter (DMM).</u>	<u>Voltage reading must be positive relative to the designated terminals.</u> <u>Reversing polarity risks catastrophic inverter failure.</u>
<u>Open Circuit Voltage (V_{oc})</u>	<u>Measured at the string ends with a DMM before connection to the combiner bus.</u>	<u>Measured V_{oc} must fall within $\pm 5\%$ of the theoretical value, adjusted for module temperature and irradiance.</u>
<u>Short Circuit Current (I_{sc})</u>	<u>Measured using a specialized PV tester or clamp meter under known irradiance conditions.</u>	<u>Measured I_{sc} must fall within $\pm 5\%$ of the irradiance-corrected theoretical value.</u>
<u>Insulation Resistance (IR)</u>	<u>Method 1 (IEC 62446-1): Short the positive and negative poles together and apply 1500V DC to earth using a Megger.</u>	<u>Absolute minimum of $1M\Omega$, though utility-scale projects typically mandate $> 50M\Omega$.</u>
<u>String Continuity</u>	<u>Confirmed indirectly via the V_{oc} and I_{sc} tests, ensuring no open circuits exist in the home-run cables.</u>	<u>Steady voltage and current readings matching expected values.</u>
<u>Earthing Continuity</u>	<u>A micro-ohmmeter is used to inject current between the PV module frames and the</u>	<u>Resistance must be $< 1\Omega$ to ensure fault currents safely dissipate, mitigating touch</u>

	<u>main earthing grid.</u>	<u>potential hazards.</u>
<u>I-V Curve Tracing</u>	<u>An I-V tracer sweeps the electrical load from short-circuit to open-circuit, plotting the current-voltage relationship.</u>	<u>The Fill Factor (FF) and curve shape must match the manufacturer's nominal flash data without step deviations.</u>

String Testing Procedure: A Step-by-Step Walkthrough

The execution of DC string testing requires highly controlled environmental conditions. Tests should ideally be conducted during periods of stable irradiance (typically greater than 600 W/m^2) to ensure accurate I-V curve generation.

The technician begins by verifying that the DC disconnect switch at the combiner box is open, effectively isolating the string from the downstream inverter. Next, the technician confirms that the irradiance sensor and RTD (Resistance Temperature Detector) are positioned at the exact plane of array of the modules under test. The positive and negative home-run cables are disconnected from the fuse holders and connected directly to the I-V tracer. The device automatically performs the V_{oc} measurement, sweeps the load to capture the I-V curve, and calculates the I_{sc} and Maximum Power Point (P_{max}). Once complete, the tracer is disconnected. Before moving to the next string, the technician performs the insulation resistance test, waiting for the capacitive charge of the PV array to dissipate fully before reconnecting the string to the combiner box terminals.

Acceptance Criteria for a Typical String

Assuming a project utilizes 400W monocrystalline modules with a Nominal V_{oc} of 49.3V and an I_{sc} of 10.3A at Standard Test Conditions (STC), configured in strings of 28 modules: The expected String V_{oc} is calculated as $28 \times 49.3\text{V} = 1380.4\text{V}$. During field testing, this value is adjusted using the module's temperature coefficient. The measured value must remain within a $\pm 5\%$ tolerance band. Similarly, the expected String I_{sc} of 10.3A is adjusted based on real-time irradiance measured by the reference cell, requiring a $\pm 5\%$ match. Insulation resistance must strictly exceed the minimum threshold at a 1500V DC test injection.

I-V Curve Anomaly Diagnostics

The I-V curve provides a highly sensitive diagnostic signature. Deviations from the

predicted curve isolate specific physical defects:

- **Notches or Steps in the Curve:** This indicates the activation of bypass diodes. It is almost exclusively caused by localized shading, severe targeted soiling (e.g., bird droppings), or a heavily fractured cell.
- **Decreased Slope near V_{oc} :** A shallower slope at the voltage axis indicates a high Series Resistance (R_s). This is typically a symptom of degraded solder bonds, poor MC4 crimps, or highly corroded connections.
- **Steeper Slope near I_{sc} :** A steeper drop-off near the current axis indicates a low Shunt Resistance (R_{sh}). This points to current leaking across the p-n junction, often a result of Potential Induced Degradation (PID) or severe manufacturing impurities in the silicon wafer.

DC Energisation Sequence

The energisation of the DC system is conducted via a bottom-up sequence, carefully closing the circuit toward the inverter to ensure safety boundaries are maintained:

1. Verify all inverter DC disconnects and combiner box load-break switches are locked open.
2. Insert all string fuses (positive and negative) at the combiner boxes, verifying polarity once more before seating the fuses.
3. Remove the LOTO on the combiner box and close the main DC disconnect switch, energising the main DC feeder cables up to the inverter station.
4. Measure the voltage at the inverter's DC input busboards to confirm the arrival of power.
5. The inverter DC switch remains open until the AC grid power is stabilized and the inverter is ready for its internal pre-charge and MPPT synchronization sequence.

3. Inverter Commissioning

The central or string inverters serve as the intelligent core of the PV plant, converting DC power to synchronized AC power while adhering to safety standards such as IEC 62109-1 and grid interaction standards.

Inverter Commissioning Sequence and Pre-Energisation Checks

Before any DC power is applied to the inverter, the T&C engineer must conduct rigorous mechanical and electrical checks. All AC and DC cable terminations must be torqued to OEM specifications, marked with torque seal, and visually inspected. The internal insulation resistance of the DC and AC busbars must be measured and recorded. Furthermore, before the inverter's AC output can be connected to the MV network, auxiliary power (typically 400V AC supplied from a local station service transformer) must be established. This powers the inverter's internal control circuitry, cooling fans, and SCADA communication modules. The engineer must verify the phase rotation

(L1-L2-L3) at the inverter AC terminals to ensure it perfectly matches the grid phase sequence. A phase reversal would result in an immediate and catastrophic fault upon synchronization.

Inverter Parameter Configuration

Once control power is established, the engineer accesses the inverter's human-machine interface (HMI) to configure the operational parameters. These settings define the inverter's behavior in relation to the UzbekEnergogrid code.

- **Grid Code Selection:** The regional grid code is selected, which establishes the baseline voltage (e.g., 800V) and frequency (50 Hz) limits.
- **Protection Settings:** Over-voltage (OV) and under-voltage (UV) trip thresholds are configured (e.g., trip instantaneously at 1.20 p.u., or ride-through for 2 seconds at 0.85 p.u.). Over-frequency (OF) and under-frequency (UF) thresholds are similarly set.
- **Active Power Curtailment:** Ramp rates are configured (e.g., 10% of nominal power per minute) to prevent sudden surges on the grid. Frequency-Watt droop settings are enabled to automatically curtail active power if the grid frequency rises dangerously high (Limited Frequency-Responsive Mode).
- **Reactive Power Control:** The Volt-VAr curve is programmed, directing the inverter to inject reactive power (capacitive) if grid voltage drops, and absorb reactive power (inductive) if grid voltage spikes.

MPPT Verification and Anti-Islanding Testing

- **Maximum Power Point Tracking (MPPT) Verification:** After closing the AC and DC breakers, the inverter initiates its start-up sequence. The engineer monitors the DC voltage and current traces on the HMI. The MPPT algorithm should actively sweep the voltage and lock onto the V_{mp} (Maximum Power Point Voltage), stabilizing the DC voltage while maximizing the AC active power output. Hunting or severe voltage oscillation indicates a tuning issue in the MPPT firmware.
- **Anti-Islanding Test:** Governed by IEC 62116, this test proves the inverter will safely shut down if it loses connection to the utility grid, preventing it from inadvertently powering a localized "island" of the grid (which poses severe risks to utility workers). The test is performed by exporting power to the grid, and then intentionally tripping the upstream MV circuit breaker. Using a high-speed power analyzer or oscilloscope, the engineer measures the time from the breaker opening to the cessation of inverter current injection. The inverter must cease energising the circuit within a specified limit, typically less than 2.0 seconds.

4. MV Ring Main Unit (RMU) Commissioning

The 33kV RMUs aggregate the 800V AC power (stepped up via adjacent local

transformers) from multiple inverter stations. Their commissioning ensures both dielectric integrity and mechanical reliability in accordance with IEC 62271-200.

Pre-Commissioning Tests for 33kV RMUs

Before high voltage is applied, RMUs undergo comprehensive cold testing. For gas-insulated RMUs, the SF6 gas pressure or density switches must be verified to be in the nominal operating zone. Low-pressure alarm and trip interlocks are functionally tested by simulating a pressure drop.

Mechanical operations form a significant part of the testing. Circuit breakers and load break switches (LBS) are operated both manually and electrically (often requiring a minimum of 50 operations) to verify mechanism smoothness. Interlocking schemes are strictly verified; for example, attempting to close an earth switch while the main busbar disconnector is closed must be physically blocked by the interlocking mechanism.

High-Pot (HV Withstand) Testing

Medium voltage equipment and interconnecting 33kV XLPE cables undergo dielectric withstand testing. To avoid the destructive space-charge accumulation effects associated with high-voltage DC testing on XLPE cables, Very Low Frequency (VLF) AC testing is utilized in accordance with IEC 60502-2.

A 0.1 Hz AC voltage is applied between the main conductor and the metallic screen. The test voltage is typically $3U_0$ (where U_0 is the phase-to-earth voltage, resulting in an applied voltage of approximately 57kV for a 33kV system). The voltage is held continuously for 15 to 60 minutes. The 0.1 Hz frequency is used because a standard 50 Hz test on kilometers of highly capacitive cable would require massive, impractical reactive power generators. The acceptance criterion is absolute: there must be no dielectric breakdown (flashover) or protective tripping of the test set during the holding period.

Contact Resistance (Ductor) Tests

To ensure that internal busbar joints and breaker contacts will not overheat under heavy load, contact resistance is measured using a micro-ohmmeter (Ductor). A highly stable 100A DC current is injected across the closed contacts, and the micro-volt drop is measured to calculate resistance. Typical acceptance values for a 33kV RMU breaker are strictly below $50\mu\Omega$. Readings exceeding this baseline indicate poor contact pressure, oxidation, or mechanical misalignment.

MV Ring Energisation Sequence

The energisation of an MV ring originates from the main 33/220kV substation and cascades outward.

1. All RMU main incoming/outgoing load break switches and local inverter MV breakers in the ring are verified to be in the OPEN position.
2. The substation's 33kV feeder breaker is closed, energising the first segment of

the underground cable up to the first RMU.

3. The T&C engineer verifies the presence of voltage using the RMU's capacitive Voltage Detecting Systems (VDS). Phase rotation is confirmed.
4. The sequence progresses outward: the outgoing switch on RMU 1 is closed, energising the cable to RMU 2. This step-by-step process continues around the ring.
5. If the ring is designed to operate as a closed loop, phasing is rigorously checked across the final open point (using a high-voltage phasing stick) before the final switch is closed. If it is an open-loop design, the designated Normal Open Point (NOP) remains locked open.

5. Substation & Protection T&C (Critical Path)

The 33/220kV grid substation is the central nervous system of the power plant. A failure during the commissioning of the main power transformer or the 220kV GIS compromises the entire facility and poses a direct threat to the stability of the UzbekEnergO transmission grid. Consequently, the testing density here is exhaustive.

Transformer Pre-Commissioning Tests

The main step-up transformer undergoes a battery of tests dictated by IEC 60076 to verify its electromagnetic, dielectric, and mechanical integrity.

<u>Test Description</u>	<u>Physical Mechanism & Procedure</u>	<u>Acceptance Criteria</u>
<u>Turns Ratio Test (TTR)</u>	<u>Compares the primary and secondary winding voltages across all positions of the On-Load Tap Changer (OLTC) to confirm the physical turns match the design ratio.</u>	<u>Measured ratio must match the nameplate ratio within $\pm 0.5\%$.</u>
<u>Winding Resistance</u>	<u>A DC current is injected into the windings until saturation occurs, measuring the resistance of the copper/aluminum conductors to identify loose joints or</u>	<u>Phase-to-phase variations must fall within 1% to 2%.</u>

	<u>broken strands.</u>	
<u>Insulation Resistance & PI</u>	<u>Measures the dielectric integrity of the cellulose paper and oil insulation by applying 5kV DC. The Polarization Index (PI) is the ratio of the 10-minute to 1-minute IR values.</u>	<u>PI > 2.0 indicates clean, dry insulation. Values < 1.0 indicate severe moisture ingress.</u>
<u>Magnetising Current Test</u>	<u>Applies a low AC voltage to the high-voltage winding while leaving the low-voltage winding open, measuring the current required to magnetize the core.</u>	<u>Compares phases; the two outer phases should draw similar current, while the center phase draws slightly less due to shorter magnetic path length.</u>
<u>Vector Group Verification</u>	<u>Applies a three-phase low voltage to the primary and measures the secondary to confirm the phase shift (e.g., YNd11 denotes a 30-degree phase shift).</u>	<u>Phase angle measurements must perfectly match the specified vector group to ensure safe parallel grid operation.</u>
<u>Sweep Frequency Response Analysis (SFRA)</u>	<u>Injects a low-voltage sinusoidal signal from 20 Hz to 1 MHz into one end of a winding and measures the response at the other. It acts as an electromagnetic "fingerprint" of the transformer's physical geometry.</u>	<u>Low frequencies (20 Hz - 2 kHz) detect core defects; Mid-frequencies (2 kHz - 20 kHz) detect bulk winding displacement; High frequencies (>20 kHz) detect localized deformation.</u>

<u>Dissolved Gas Analysis (DGA)</u>	<u>Extracts and analyzes gases dissolved in the insulating oil using gas chromatography, identifying fault signatures based on the thermal breakdown of oil and paper.</u>	<u>Evaluated against IEEE C57.104 standard limits (detailed below).</u>
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Deep Dive: Dissolved Gas Analysis (DGA) Interpretation DGA is the most powerful predictive diagnostic tool available to a T&C Manager. Internal faults produce specific gases depending on the energy and temperature of the fault.

- **Hydrogen (H_2)**: Generated by low-energy corona or partial discharge.
- **Methane (CH_4) and Ethane (C_2H_6)**: Indicate low-to-medium temperature thermal faults in the oil ($150^\circ C - 300^\circ C$).
- **Ethylene (C_2H_4)**: Indicates severe, high-temperature thermal faults exceeding $300^\circ C$, often linked to localized hot spots on the core or bad connections.
- **Acetylene (C_2H_2)**: This gas requires immense energy to form (temperatures $> 700^\circ C$) and is the definitive signature of active arcing or high-energy electrical discharge. Any detectable level of Acetylene in the main tank warrants an immediate halt to energisation and internal investigation.
- **Carbon Monoxide (CO) & Carbon Dioxide (CO_2)**: While formed during normal oil aging, rapidly elevated levels specifically indicate the thermal degradation of the solid cellulose paper insulation.

GIS Pre-Commissioning Tests

The 220kV Gas Insulated Switchgear (GIS) requires highly specialized testing per IEC 62271-203.

- **SF6 Gas Quality and Leakage**: The SF6 gas must be verified for purity ($>97\%$). Crucially, the dew point must be $\leq -36^\circ C$ (approx. 200 ppmv moisture). If moisture levels are too high, arcing during breaker operation will hydrolyze the SF6, creating highly corrosive and toxic hydrofluoric acid (HF). Leakage is tested using a helium mass spectrometer; the leakage rate must be strictly $<0.5\%$ (and often $<0.1\%$ per utility specification) per year.
- **High-Pot (Power Frequency Withstand) and UHF PD**: A resonant test set applies an elevated AC voltage (e.g., 460kV for 1 minute) to the fully assembled GIS to expose any microscopic defects or mobile metallic particles. Concurrently, Ultra-High Frequency (UHF) sensors monitor for Partial Discharge (PD).

Acceptance requires PD levels to remain strictly below 5 pC.

Protection Relay Testing: The Secondary Injection Process

Protection relays are intelligent electronic devices (IEDs) that monitor the power system and issue rapid trip commands during faults. They are tested using advanced secondary injection test sets (such as the OMICRON CMC 356) which simulate the low-voltage, low-current signals normally provided by the substation's Current Transformers (CTs) and Voltage Transformers (VT).

1. **Distance Relay (21):** Protects the 220kV transmission line by calculating the impedance ($Z = V/I$) to the fault. The test set injects specific voltage and current phasors to simulate faults at varying distances and phase angles. The relay is tested to verify it trips instantaneously for faults in Zone 1 (typically 80% of the line length) and trips with a time delay for Zone 2 (typically 120% of the line). The engineer plots these tripping points to ensure they form the correct quadrilateral or mho circle characteristic.
2. **Differential Relay (87T for Transformer, 87B for Busbar):** The 87T relay compares the current entering the transformer's 220kV side against the current leaving the 33kV side. The test verifies the "dual-slope" characteristic, ensuring the relay remains stable during minor mismatches caused by tap changer operation. Crucially, the test set injects 2nd and 5th harmonic currents to verify the "harmonic restraint" feature, which prevents the relay from falsely tripping during the massive inrush current that occurs when a transformer is first energized.
3. **Overcurrent (51/50) and Earth Fault (51N/50N):** The test set injects gradually increasing currents to verify the exact "pickup" threshold where the relay recognizes a fault. It then injects high fault currents to verify the operating time matches the configured Inverse Definite Minimum Time (IDMT) curves (e.g., IEC Standard Inverse).
4. **End-to-End Protection Test:** To verify teleprotection schemes (such as Permissive Overreach Transfer Trip - POTT), two GPS-synchronized test sets are placed at opposite ends of the 220kV line. Simulated faults are injected simultaneously at exactly the same microsecond to verify that the relays communicate via fiber optics and issue the correct coordinated trip commands.

Protection Settings Approval Process

Before any relays can be put into service, a comprehensive Protection Coordination Study is submitted to UzbekEnergo. This study models the entire grid and calculates the exact settings required to ensure that a fault in the solar plant is cleared locally without tripping the broader utility grid. UzbekEnergo reviews the relay setting files. Once approved, the utility issues a formal consent document, and the settings are locked into the IEDs. Any subsequent changes require formal dispensation.

Substation Energisation Sequence (Dead Bus to Live 220kV)

The energisation of the 220kV substation is a meticulously scripted and tension-filled operation.

1. **Verification of Isolation and LOTO Clearance:** The T&C Manager verifies that all breakers, disconnectors, and earth switches in the substation are open. All Permits to Work are surrendered, LOTO locks are removed, and the site is declared a high-voltage live zone.
2. **Protection Readiness:** Engineers verify that all relay self-monitoring alarms are clear, trip circuit supervision is healthy, and the 87T differential protection is armed. The transformer tap changer is locked in the nominal design position.
3. **Busbar Energisation:** Coordinating via radio with the utility dispatcher, the utility end of the 220kV line is closed. The local substation line disconnectors are closed. The T&C Manager commands the closing of the main 220kV incomer breaker. The 220kV GIS busbar is now energized. Voltage presence and phase sequence are immediately verified via the HMI utilizing the VT secondary voltages.
4. **Transformer Energisation:** With the busbar live, the T&C Manager commands the closing of the 220kV transformer breaker. The transformer undergoes immense electromagnetic forces, emitting a loud, deep hum (magnetostriction). Protection engineers stare intensely at the relay HMIs, monitoring the inrush current decay and ensuring the harmonic restraint prevents a spurious 87T trip.
5. **The Soak Test:** The transformer is left energized at no-load for a mandatory "soak" period of 24 hours. This allows any localized gas bubbles trapped in the insulating oil to dissolve and allows engineers to monitor for steady-state thermal anomalies or partial discharge acoustics.
6. **MV Energisation:** Following a successful soak, the transformer's 33kV secondary breaker is closed, energising the MV busbar. Phasing is verified across the busbar before any downstream RMU feeders are permitted to be energized.

6. Grid Connection & SCADA

Grid Connection Application and Utility Approval

The grid connection process with UzbekEnergo requires a robust administrative and technical submission. Documents submitted include the Grid Impact Study, as-built Single Line Diagrams (SLDs), protection coordination studies, and all certified pre-commissioning test reports. Utility energisation approval is granted only after UzbekEnergo engineers physically witness key tests, typically including the main transformer differential testing and the end-to-end teleprotection validation.

Synchronisation Test

Once the utility grants consent, the synchronisation test is performed on the first block of inverters. When commanded to start, the inverter samples the utility grid voltage and frequency via its AC terminals. It adjusts the firing sequence of its internal Insulated-Gate Bipolar Transistors (IGBTs) to generate a waveform that perfectly matches the grid's phase angle and frequency. Once synchronized, the inverter closes its internal AC contactor. The SCADA engineer monitors the seamless ramp-up of active power injection to ensure there are no transient spikes or reactive power oscillations.

Grid Code Compliance Test (Uzbekistan)

The HELIOS project must demonstrate dynamic compliance with the Grid Code of Uzbekistan and international interoperability standards like IEEE 2800.

- **Reactive Power Capability:** The plant is commanded to operate at varying power factors (e.g., 0.95 leading to 0.95 lagging). The utility monitors the Point of Common Coupling (PCC) to ensure the plant can successfully absorb or inject reactive power to regulate grid voltage.
- **Frequency Response (LFRM):** The plant's ability to operate in Limited Frequency-Responsive Mode is tested. If the grid frequency exceeds a specified deadband, the plant must automatically curtail active power output following a defined droop characteristic to help stabilize the grid.
- **Fault Ride-Through (FRT):** Simulated models and hardware-in-the-loop tests demonstrate that the inverters will not trip during transient grid voltage sags (Low Voltage Ride-Through) or surges, instead providing stabilizing reactive currents to aid grid recovery.

SCADA Commissioning

The Supervisory Control and Data Acquisition (SCADA) system is the central monitoring hub, utilizing the IEC 61850 protocol standard for utility automation. During commissioning, the SCADA engineer configures the network architecture and tests communication pathways to all IEDs and inverters. Data points are mapped using standardized Logical Nodes (LN) per the IEC 61850-7-420 Distributed Energy Resources extension.

- **ZINV:** Represents the Inverter operational state.
- **DPVA:** Represents the PV Array DC characteristics.
- **MMXU:** Represents metered operational values (voltage, current, active/reactive power).
- **XCBR:** Represents Circuit Breaker status and controls. The engineer executes point-to-point testing, commanding breakers to open/close from the HMI and verifying that field alarms populate the event list with accurate timestamps.

Performance Testing

- **Performance Ratio (PR) Test:** The critical commercial metric for the EPC

contractor. PR is the ratio of actual energy exported to the grid versus the theoretical energy the plant should have produced based on measured weather conditions. The test is typically run over a continuous 7 to 30-day period. High-precision pyranometers measure irradiance, and RTDs measure module temperature. The acceptance threshold is heavily negotiated but is typically defined at $>80\%$.

- **Energy Yield Test:** A long-term measurement, often spanning the first year of operation, comparing the absolute MWh generated against the financial P50/P90 models predicting plant profitability.

7. Commissioning to Handover

The conclusion of physical testing initiates the administrative handover of the asset from the EPC contractor to the client.

- **Provisional Acceptance Certificate (PAC):** Issued when the plant has successfully passed the PR test, is safely connected to the grid, and is capable of continuous power export. The client, utility, and EPC representatives witness the final performance runs. The issuance of the PAC is a major milestone, marking the transfer of operational risk and asset ownership to the client.
- **Taking-Over Certificate:** Often used interchangeably with the PAC, it formally certifies that the contractor has completed the core works, subject to a finalized punch list of minor, non-critical defects.
- **Defects Notification Period (DNP):** A warranty period, typically lasting 12 to 24 months post-PAC. During the DNP, the EPC contractor remains contractually liable for rectifying any equipment failures, SCADA glitches, or performance shortfalls that arise.
- **Final Acceptance Certificate (FAC):** Issued at the culmination of the DNP, signifying that all punch list items are closed, the plant has performed reliably, and the final retention payments can be released to the contractor.
- **O&M Training:** Prior to handover, the T&C team provides the client's Operations & Maintenance (O&M) personnel with deep-dive training. This includes SCADA architecture navigation, LOTO and isolation procedures, and advanced troubleshooting for inverter and relay faults.
- **As-Built Documentation:** The T&C team hands over a comprehensive turnover package. This includes red-lined as-built drawings reflecting all field modifications, OEM manuals, complete signed test reports, protection relay setting files, and software backups.

8. Real Document Templates

The following templates represent the rigorous, structured data formats utilized by T&C engineers in the field.

8.1 Commissioning Plan Structure

<u>Section</u>	<u>Title</u>	<u>Detailed Contents</u>
<u>1.0</u>	<u>Project Overview & Scope</u>	<u>Plant description, single line diagram boundaries, applicable codes (IEC, IEEE, Grid Code).</u>
<u>2.0</u>	<u>HSE & LOTO Management</u>	<u>Permit to Work (PTW), Sanction for Test (SFT), isolation protocols, switching authorization levels.</u>
<u>3.0</u>	<u>Organization & Responsibilities</u>	<u>T&C Manager, Lead Protection Engineer, SCADA Engineer, Utility Interfaces, Client Witnesses.</u>
<u>4.0</u>	<u>Pre-Commissioning Procedures</u>	<u>Methodologies for DC string testing, MV Hi-pot, Transformer TTR/DGA, Relay secondary injection.</u>
<u>5.0</u>	<u>Energisation Sequence</u>	<u>Step-by-step switching programs: 220kV Substation → 33kV MV Ring → Inverters.</u>
<u>6.0</u>	<u>Grid Compliance & Perf. Testing</u>	<u>PR Test methodology, Fault Ride-Through (FRT) demonstration, reactive power capability test.</u>
<u>7.0</u>	<u>Handover & Documentation</u>	<u>PAC criteria, punch list management, O&M training schedule, as-built turnover index.</u>

8.2 DC System Pre-Commissioning Checklist (String Level)

<u>Item</u>	<u>Verification Step</u>	<u>Methodology / Standard</u>	<u>Acceptance Criteria</u>	<u>Result</u>
<u>1</u>	<u>Visual Inspection & Torque</u>	<u>Check MC4 seating: torque to OEM spec</u>	<u>No damage; Torque marks present</u>	<u>[Pass]</u>
<u>2</u>	<u>Polarity Check</u>	<u>DMM across pos/neg at combiner box</u>	<u>Positive voltage matching terminal</u>	<u>[Pass]</u>
<u>3</u>	<u>Open Circuit Voltage (V_{oc})</u>	<u>DMM measurement</u>	<u>$\pm 5\%$ of temperature-corrected STC</u>	<u>[Pass]</u>
<u>4</u>	<u>Short Circuit Current (I_{sc})</u>	<u>PV clamp meter / tracer</u>	<u>$\pm 5\%$ of irradiance-corrected STC</u>	<u>[Pass]</u>
<u>5</u>	<u>Insulation Resistance (IR)</u>	<u>IEC 62446-1 (Method 1) at 1500V DC</u>	<u>> 50MΩ to earth</u>	<u>[Pass]</u>
<u>6</u>	<u>Earthing Continuity</u>	<u>Micro-ohmmeter injection</u>	<u>< 1Ω between frame and grid</u>	<u>[Pass]</u>

8.3 String IV Curve Test Record

Project: HELIOS | Module Type: 400W Mono | String Config: 28 in series

<u>String ID</u>	<u>Irradiance (W/m2)</u>	<u>Temp (°C)</u>	<u>Meas. Voc (V)</u>	<u>Meas. Isc (A)</u>	<u>Pmax (kW)</u>	<u>Fill Factor</u>	<u>Status / Diagnostics</u>
<u>CB04-S10</u>	<u>850</u>	<u>45.2</u>	<u>1310.5</u>	<u>8.85</u>	<u>8.92</u>	<u>0.77</u>	Pass
<u>CB04-S11</u>	<u>845</u>	<u>45.4</u>	<u>1308.2</u>	<u>8.81</u>	<u>8.87</u>	<u>0.76</u>	Pass
<u>CB04-S12</u>	<u>850</u>	<u>45.3</u>	<u>1150.1</u>	<u>8.82</u>	<u>7.65</u>	<u>0.75</u>	Fail (<u>Curve step: bypass diode / shading</u>)
<u>CB04-S13</u>	<u>848</u>	<u>45.1</u>	<u>1309.0</u>	<u>8.80</u>	<u>8.10</u>	<u>0.68</u>	Fail (<u>Low FF: High series resistance/bad crimp</u>)

8.4 Inverter Commissioning Record

<u>Parameter / Check</u>	<u>Configured Value / Method</u>	<u>Acceptance / Status</u>
<u>Inverter ID</u>	<u>INV-STATION-12</u>	<u>N/A</u>
<u>AC Phase Rotation</u>	<u>Measured L1-L2-L3</u>	<u>Matches Grid (Clockwise)</u>

<u>DC Insulation Resistance</u>	<u>Internal self-test</u>	<u>> 100kΩ warning threshold cleared</u>
<u>Grid Code Selection</u>	<u>UZBEKENERGO_HV</u>	<u>Verified on HMI</u>
<u>Over-Voltage Trip</u>	<u>1.15 p.u. (delay 0.5s)</u>	<u>Setting verified</u>
<u>Under-Voltage Ride Through</u>	<u>0.85 p.u. (delay 2.0s)</u>	<u>Setting verified</u>
<u>Anti-Islanding Test (IEC 62116)</u>	<u>Trip upstream 33kV breaker</u>	<u>Disconnection in 1.2s (Limit < 2.0s)</u>
<u>MPPT Tracking Stability</u>	<u>Observation under load</u>	<u>Stable lock on V_{mp}; no hunting</u>

8.5 Transformer Pre-Commissioning Test Report

<u>Test Description</u>	<u>Reference Standard</u>	<u>Measured Value</u>	<u>Acceptance Criteria</u>	<u>Result</u>
<u>Turns Ratio (Tap Nom)</u>	<u>IEC 60076</u>	<u>1.002 (A-a)</u>	<u>$\pm 0.5\%$ of nameplate ratio</u>	<u>Pass</u>
<u>Winding Resistance</u>	<u>IEC 60076</u>	<u>Max Var: 1.1%</u>	<u>$< 2.0\%$ variation between phases</u>	<u>Pass</u>

<u>Insulation (PI)</u>	<u>IEEE 43</u>	<u>2.45</u>	<u>PI > 2.0</u>	<u>Pass</u>
<u>DGA:</u> <u>Hydrogen (H_2)</u>	<u>IEEE</u> <u>C57.104</u>	<u>12 ppm</u>	<u>< 100 ppm</u> <u>(Condition 1)</u>	<u>Pass</u>
<u>DGA:</u> <u>Acetylene (C_2H_2)</u>	<u>IEEE</u> <u>C57.104</u>	<u>0 ppm</u>	<u>strictly < 1 ppm</u>	<u>Pass</u>
<u>Vector Group</u>	<u>IEC 60076</u>	<u>YNd11</u>	<u>Matches</u> <u>30-degree phase</u> <u>shift</u>	<u>Pass</u>

8.6 Protection Relay Test Record (Overcurrent 51/50)

<u>Relay ID</u>	<u>Make/Model</u>	<u>CT Ratio</u>	<u>Characteristic</u>	<u>Pickup Setting (Is)</u>	<u>Time Multiplier (TMS)</u>
<u>RLY-3</u> <u>3-FDR</u> <u>1</u>	<u>SEL-751</u>	<u>1000/1</u> <u>A</u>	<u>IEC Standard</u> <u>Inverse</u>	<u>1.20 A</u> <u>(1200 A</u> <u>primary)</u>	<u>0.15</u>

Injection Test Results (Phase A):

<u>Injected Current</u>	<u>Expected Trip Time (s)</u>	<u>Actual Trip Time (s)</u>	<u>Deviation (%)</u>	<u>Result</u>
<u>$2.0 \times I_s$ (2.4</u>	<u>1.504</u>	<u>1.512</u>	<u>+0.53%</u>	<u>Pass (Limit</u>

A)				$\pm 5\%$
$5.0 \times I_s$ (6.0 A)	0.642	0.640	-0.31%	Pass
$10.0 \times I_s$ (12.0 A)	0.446	0.448	+0.44%	Pass

8.7 SCADA Point List (IEC 61850 Logical Nodes)

<u>Signal Description</u>	<u>IEC 61850 Node</u>	<u>Data Attribute</u>	<u>Data Type</u>	<u>IEC 61850 Object Path</u>
<u>Inverter Active Power</u>	MMXU	TotW.mag.f	Analog	INV1Ctrl/MMXU1.TotW.mag.f
<u>Inverter Grid Voltage</u>	MMXU	PhV.phsA.cVal	Analog	INV1Ctrl/MMXU1.PhV.phsA.cVal
<u>Inverter Operating State</u>	ZINV	OpSt.stVal	Enum	INV1Ctrl/ZINV1.OpSt.stVal
<u>PV Array DC Voltage</u>	DPVA	Vol.mag.f	Analog	INV1Ctrl/DPVA1.Vol.mag.f

<u>Main Breaker Status</u>	<u>XCBR</u>	<u>Pos.stVal</u>	<u>Double Point</u>	<u>SUB1Ctrl/XCBR1.Pos.stVal</u>
<u>Transformer Temp Alarm</u>	<u>PTTR</u>	<u>AlmThm.stVal</u>	<u>Boolean</u>	<u>SUB1Ctrl/PTTR1.AlmThm.stVal</u>

8.8 Punch List (T&C Category A - Critical)

<u>Punch ID</u>	<u>Location</u>	<u>Description</u>	<u>Category</u>	<u>Action Required</u>	<u>Status</u>
<u>PL-001</u>	<u>INV-05</u>	<u>Phase A AC termination missing torque mark.</u>	<u>A (Pre-Energisation)</u>	<u>Retorque and mark.</u>	<u>Open</u>
<u>PL-002</u>	<u>RMU-12</u>	<u>SF6 pressure gauge indicates low warning.</u>	<u>A (Pre-Energisation)</u>	<u>Top up SF6; check for leaks.</u>	<u>Open</u>
<u>PL-003</u>	<u>SCADA</u>	<u>Comm failure to Met Station 2.</u>	<u>B (Pre-PAC)</u>	<u>Check fiber optic terminations.</u>	<u>Closed</u>

8.9 Provisional Acceptance Certificate (PAC) Format

PROVISIONAL ACCEPTANCE CERTIFICATE (PAC) **Project:** HELIOS 100 MW Solar PV Plant **Contractor:** [EPC Name] | **Client:** [Owner Name] **Date of Issuance:** [Date]

This certificate confirms that the HELIOS 100 MW Solar PV Plant has successfully completed all required pre-commissioning, commissioning, and performance testing, including the Performance Ratio (PR) test, achieving a PR of [XX.X]% against the guaranteed threshold of 80.0%. The facility is synchronized with the UzbekEnergogrid and is capable of continuous, safe commercial operation.

Operational control and risk of loss are hereby transferred to the Client. The Contractor remains liable for the rectification of the attached Punch List items and all obligations under the Defects Notification Period (DNP) commencing on the date of this certificate.

Signatures: [T&C Manager] _____ [Owner's Engineer] _____
[Utility Rep] _____

8.10 T&C Handover Document Index

<u>Index</u>	<u>Document Title</u>	<u>Format / Type</u>
<u>1.0</u>	<u>As-Built Single Line Diagrams (SLD)</u>	<u>PDF / DWG</u>
<u>2.0</u>	<u>Substation Test Reports (TTR, DGA, SFRA, GIS)</u>	<u>PDF (Signed)</u>
<u>3.0</u>	<u>Protection Relay Setting Files & Test Reports</u>	<u>.rdb / .set / PDF</u>
<u>4.0</u>	<u>DC String IV Curve Test Database</u>	<u>Excel / PDF</u>
<u>5.0</u>	<u>Grid Compliance & PR Test Final Report</u>	<u>PDF</u>
<u>6.0</u>	<u>OEM Operation and Maintenance Manuals</u>	<u>PDF</u>

9. T&C Interfaces

The execution of commissioning is highly interdisciplinary, requiring seamless interfaces with multiple project stakeholders.

- **With Construction (SWITCHMAN):** The handover from construction to T&C is formalized via Walkdown and Mechanical Completion (MC) certificates. The

handover package must include red-lined drawings, QA/QC torque records, and confirmation that all debris and scaffolding have been removed from electrical clearance zones.

- **With Quality (OHMMETER):** The QC department primarily witnesses static visual inspections and structural torquing. T&C carries out dynamic and electrical testing (e.g., secondary injection, Hi-pot) independently. However, QC collates and archives the final generated test reports for the turnover package.
- **With EHS (WARDEN):** During the transition to energisation, the LOTO system shifts from a construction-controlled multi-lock system to a strict T&C-controlled system. The T&C Manager becomes the ultimate Control Person, authorizing all isolations and ensuring boundary limits are respected by construction crews finishing adjacent works.
- **With Engineering (CIPHER, VOLTA):** Engineering provides the theoretical relay setting calculations and SCADA logic matrices. T&C physical uploads these settings into the IEDs, executes the tests, and provides feedback if field conditions (e.g., unanticipated CT saturation characteristics) require tuning.
- **With Utility (GRIDMASTER):** The utility mandates specific witness points to ensure grid safety. UzbekEnergo must typically be present to witness the main transformer differential testing, the end-to-end teleprotection tests, and the final synchronization sequence of the plant.
- **With the Client:** The owner's engineer interfaces continuously with T&C, witnessing key performance milestones, auditing test records, and confirming the resolution of major punch list items prior to signing off on the PAC.

10. Common T&C Problems & Resolution Strategies

1. **String IV curve shows 15% lower power than expected.**
 - **Investigation:** The technician analyzes the shape of the IV curve. If the voltage (V_{oc}) is normal but the curve exhibits a "step" or notch, it indicates a bypass diode has activated, typically pointing to localized shading, targeted soiling, or a shattered module. If the curve lacks steps but has a shallow slope near the voltage axis, it indicates high series resistance (R_s). The remedy involves a physical walk-through to check for poorly crimped MC4 connectors or degraded junction boxes.
2. **Transformer turns ratio (TTR) test fails.**
 - **Investigation:** If the measured ratio deviates by more than 0.5%, the engineer must first rule out residual magnetism (often left behind by previous DC winding resistance tests) by executing a demagnetization routine on the test set. If the test still fails, it implies a severe internal defect: shortened turns, open windings, or tap changer misalignment. The

transformer cannot be energized; the OEM must be brought to the site for internal inspection and repair.

3. **Utility rejects protection relay settings after commissioning is complete.**

- Investigation: This usually occurs if the utility performs a revised grid fault study and determines the plant's relays lack coordination with upstream substations. The T&C Manager must halt further energisation, collaborate with the engineering design team to recalculate the time multiplier settings (TMS) or distance relay reaches, submit the revised study for utility approval, and re-test the relays using secondary injection.

4. **SCADA fails to communicate with 3 inverters.**

- Investigation: The SCADA engineer initiates a layered troubleshooting process. Layer 1 (Physical): Use an OTDR on fiber optics or continuity tester on RS485 to ensure the cable isn't severed. Layer 2 (Network): Ping the IP addresses to verify subnet masks and router configurations are correct. Layer 3 (Protocol): Verify the IEC 61850 Configured IED Description (CID) files to ensure the logical nodes in the SCADA database match the actual firmware mapping in the inverters.

5. **Utility delays energisation consent by 4 weeks.**

- Impact: This immediately disrupts the critical path, delaying the PR test and commercial operation. Commercially, it exposes the EPC contractor to Liquidated Damages (LDs). The T&C Manager must meticulously document the readiness of the plant and all correspondence with the utility to support the project management team in filing an Extension of Time (EOT) claim, proving the delay was solely utility-driven.

6. **PR test fails (falls below 80% threshold).**

- Investigation: T&C assesses three primary vectors. First, they check the pyranometers for severe soiling or misalignment, which would skew the denominator (expected energy) artificially high. Second, they analyze SCADA logs for inverter curtailment events caused by grid overvoltage or utility dispatch commands, which must be factored out of the calculation. Third, they look for genuine DC side underperformance (e.g., blown fuses, PID, pervasive soiling). The remedy involves recalibrating met stations, cleaning the arrays, replacing failed components, and re-running the 7-to-30 day test.